

```

78.     # Find at least seven words that have at least one matching letter
79.     # (but give up at 500 tries if not enough can be found).
80.     for i in range(500):
81.         if len(words) == 12:
82.             break # Found 7 or more words, so break out of the loop.
83.
84.         randomWord = getOneWordExcept(words)
85.         if numMatchingLetters(secretPassword, randomWord) != 0:
86.             words.append(randomWord)
87.
88.     # Add any random words needed to get 12 words total.
89.     while len(words) < 12:
90.         randomWord = getOneWordExcept(words)
91.         words.append(randomWord)
92.
93.     assert len(words) == 12
94.     return words
95.
96.
97. def getOneWordExcept(blocklist=None):
98.     """Returns a random word from WORDS that isn't in blocklist."""
99.     if blocklist == None:
100.         blocklist = []
101.
102.     while True:
103.         randomWord = random.choice(WORDS)
104.         if randomWord not in blocklist:
105.             return randomWord
106.
107.
108. def numMatchingLetters(word1, word2):
109.     """Returns the number of matching letters in these two words."""
110.     matches = 0
111.     for i in range(len(word1)):
112.         if word1[i] == word2[i]:
113.             matches += 1
114.     return matches
115.
116.
117. def getComputerMemoryString(words):
118.     """Return a string representing the "computer memory"."""
119.
120.     # Pick one line per word to contain a word. There are 16 lines, but
121.     # they are split into two halves.
122.     linesWithWords = random.sample(range(16 * 2), len(words))
123.     # The starting memory address (this is also cosmetic).
124.     memoryAddress = 16 * random.randint(0, 4000)
125.
126.     # Create the "computer memory" string.
127.     computerMemory = [] # Will contain 16 strings, one for each line.
128.     nextWord = 0 # The index in words of the word to put into a line.
129.     for lineNum in range(16): # The "computer memory" has 16 lines.
130.         # Create a half line of garbage characters:

```